

UML DIAGRAMS

USE CASE, SEQUENCE, CLASS & RELATIONSHIPS



USE CASE Diagram

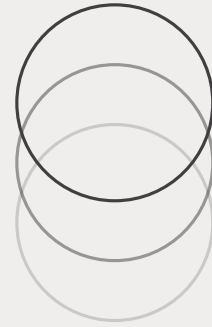
SUMMARY

A Use Case Diagram is a type of UML (Unified Modeling Language) behavioural diagram that shows who uses a system and what they can do with it. It answers two fundamental questions:

- Who interacts with this system? (Actors)
- What can they do? (Use Cases)

Think of it as a bird's-eye view of system functionality from the user's perspective — it doesn't show how things work internally, only what the system does and for whom.

USE CASE Diagram



THE 4 CORE ELEMENTS

1. Actor A stick figure representing anyone or anything that interacts with the system from outside. Actors can be:

- A person (Student, Librarian, Admin)
- An external system (Payment Gateway, Email Server)
- A device (ATM, Sensor)

Key rule: an actor is always outside the system boundary.

2. Use Case (oval) An oval/ellipse representing a specific action or function the system performs for an actor. Written as a verb phrase — e.g. "Borrow Book", "Login", "Generate Report". Each use case delivers some value to the actor.

3. System Boundary (rectangle) A box drawn around all the use cases to represent the scope of the system being modelled. Everything inside the box is part of the system. Actors always sit outside it.

4. Relationships (lines/arrows)

- Association — a plain line connecting an actor to a use case (means "this actor uses this feature")
- <<include>> — one use case always triggers another (e.g. "Make Payment" always includes "Validate Card")
- <<extend>> — one use case optionally adds behaviour to another (e.g. "Login" may extend with "Show CAPTCHA" only when too many failed attempts)
- Generalisation — one actor or use case inherits from another (like inheritance in OOP)

UML USE CASE – THREE ASSOCIATION TYPES



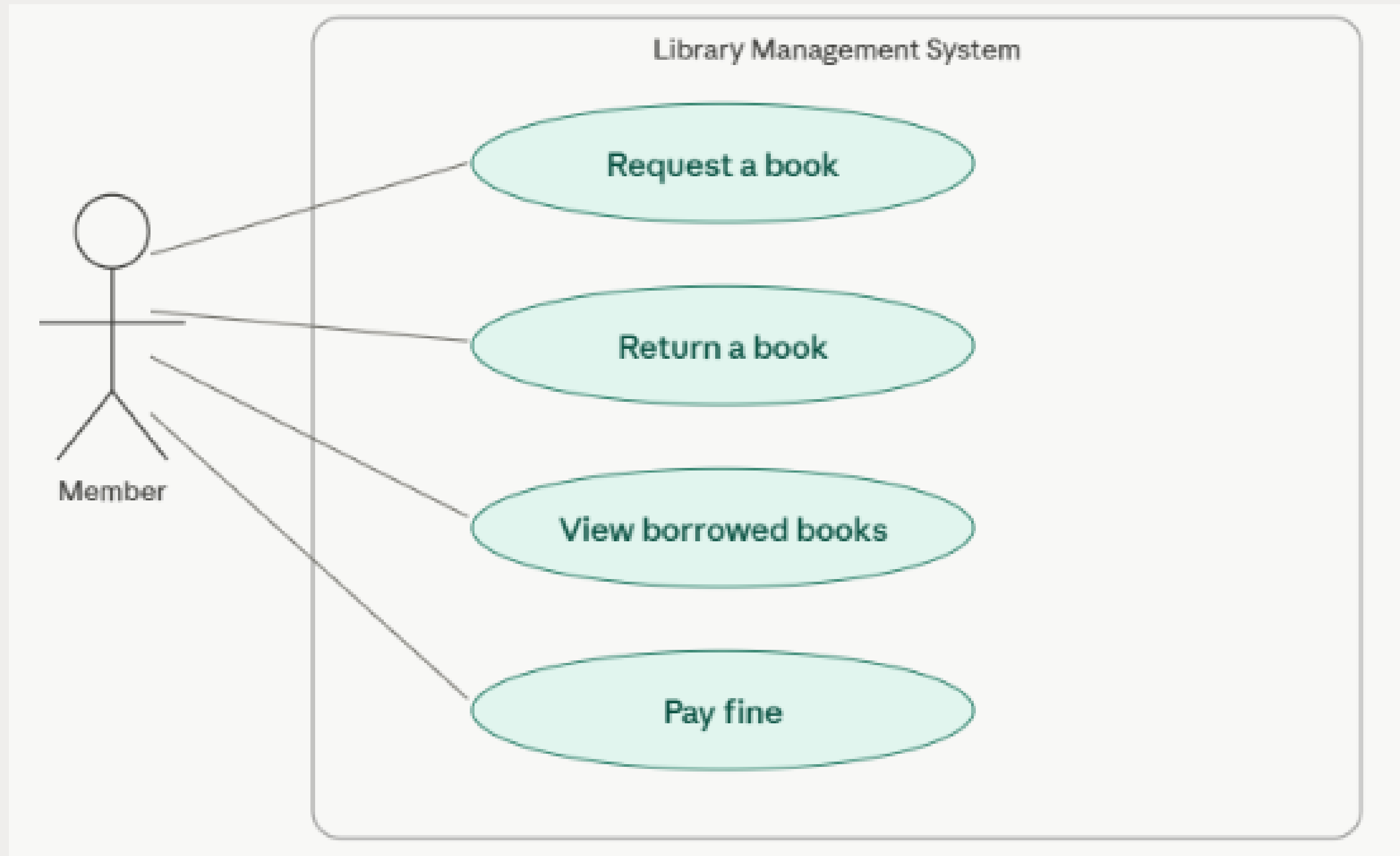
USE CASE Diagram

QUESTION

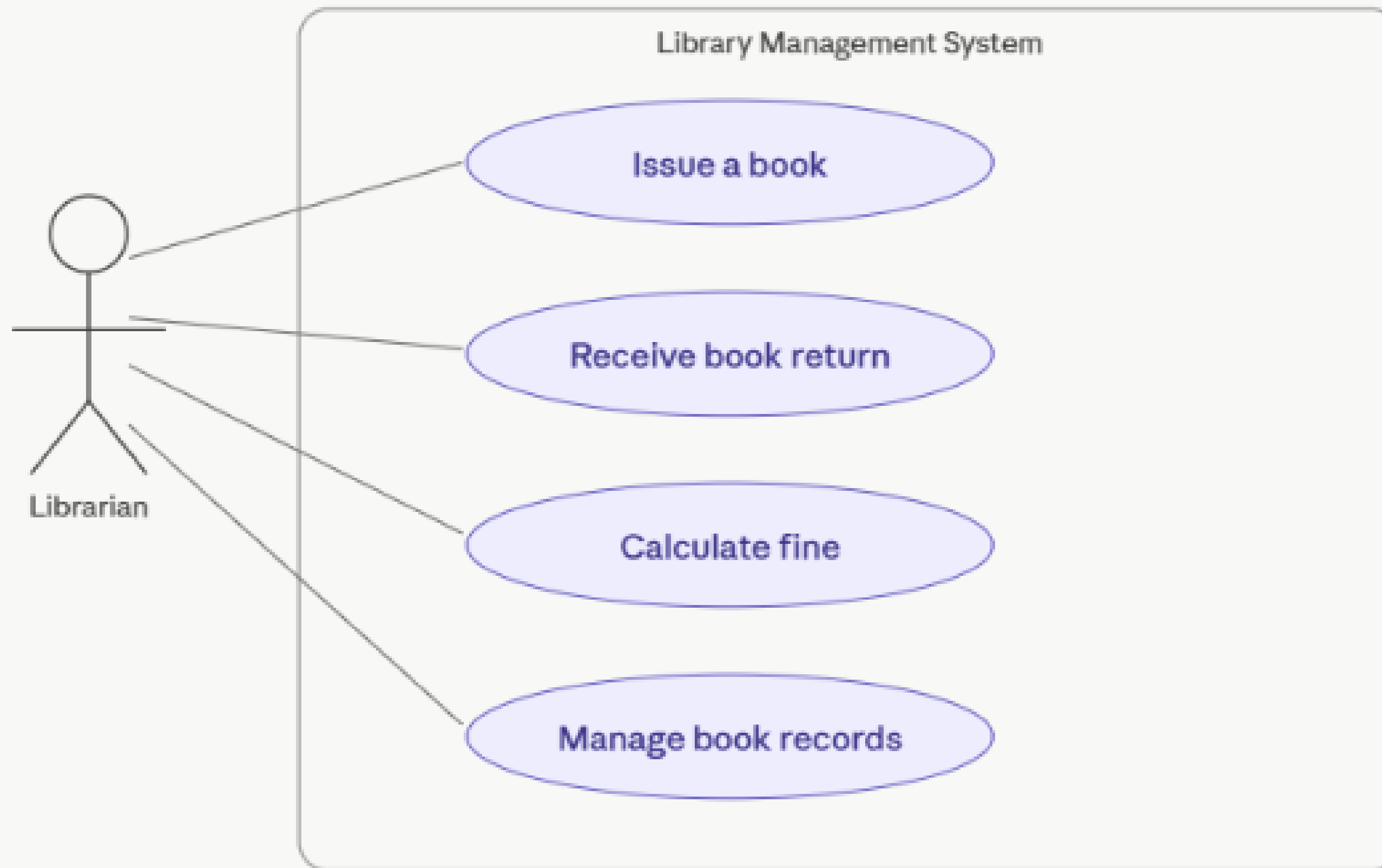
Consider a Library Management System with the following classes: **Book, Member, Librarian, Transaction, Fine.**

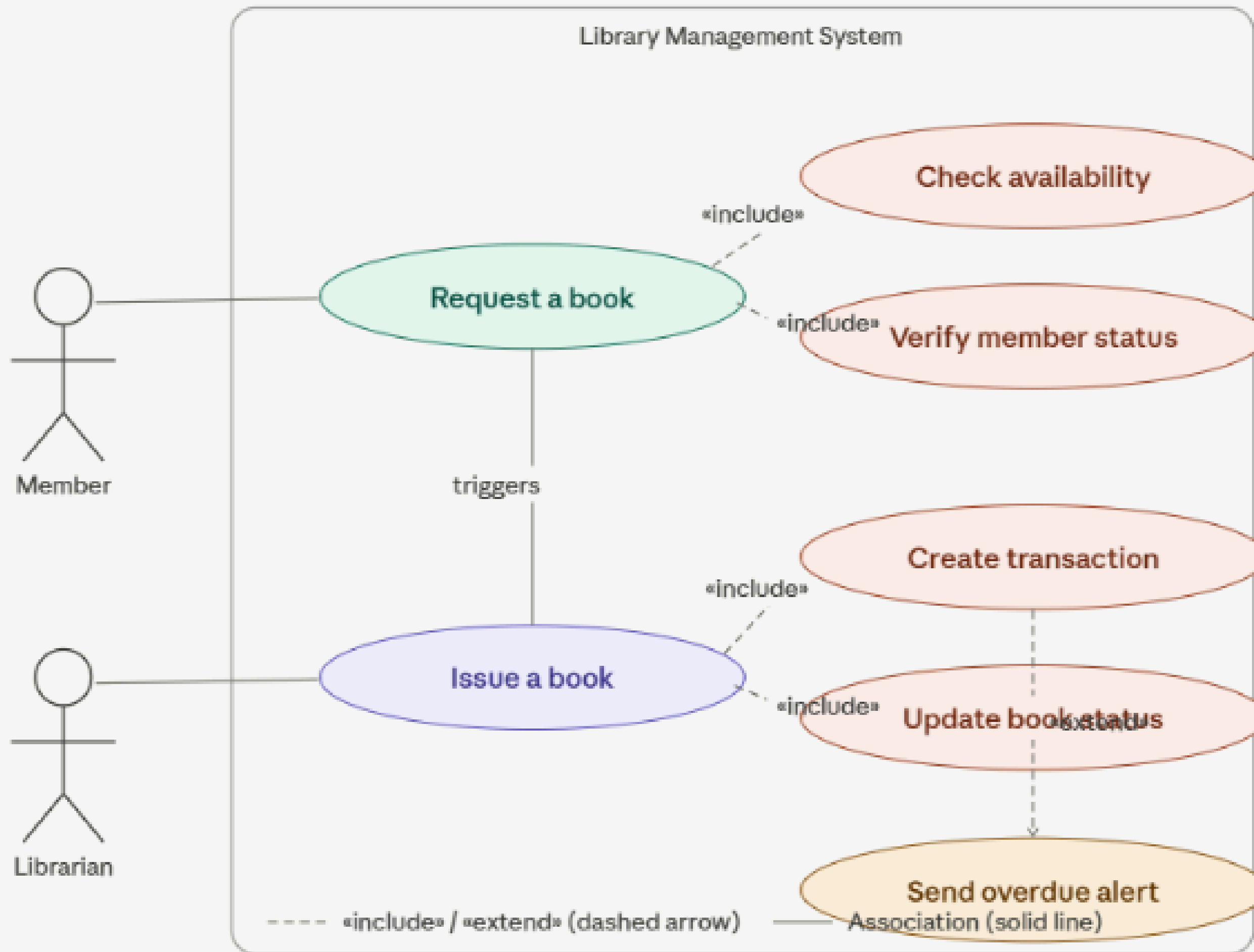
The system functions as follows:

- A Member requests to issue a book.
- The Librarian verifies the availability of the book.
- If the book is available, it is issued and a Transaction record is created.
- When the member returns the book, the system checks the return date.
- If the book is returned late, a fine is calculated and collected.

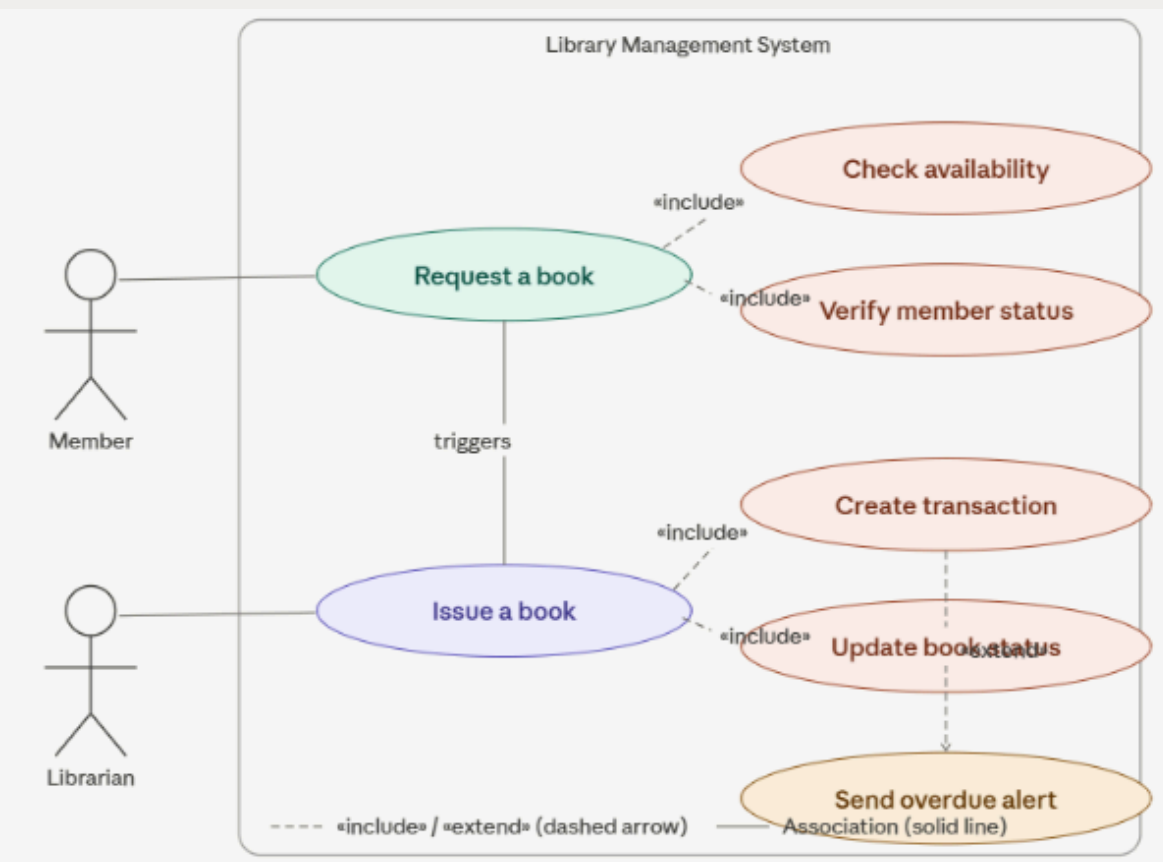


High-level use case diagram – Librarian





Relationships



SIMPLE ASSOCIATION (ACTOR ↔ USE CASE)

MEMBER ↔ REQUEST TO ISSUE BOOK

LIBRARIAN ↔ ISSUE BOOK

«INCLUDE» RELATIONSHIPS

ISSUE BOOK → CHECK BOOK AVAILABILITY

ISSUE BOOK → UPDATE BOOK STATUS

«EXTEND» RELATIONSHIP

SEND OVERDUE ALERT «EXTEND» → ISSUE BOOK



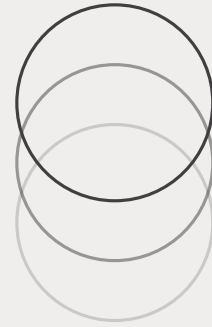
SEQUENCE Diagram

SUMMARY

A Sequence Diagram is a UML behavioural diagram that shows how objects interact with each other over time, in a specific order. While a use case diagram answers "what does the system do?", a sequence diagram answers "how and in what order do the objects talk to each other?"

It captures the flow of messages between objects from top to bottom — time moves downward.

SEQUENCE Diagram



THE 4 CORE ELEMENTS

- 1. Actor / Object (at the top)** Represented as a labelled box at the top of the diagram. Each participant in the interaction gets one box — could be a person (Customer), a system component (Order System), or a database.
- 2. Lifeline (vertical dashed line)** A dashed vertical line hanging down from each object. It represents the object's existence over time. As long as the lifeline is visible, the object is alive in the interaction.
- 3. Activation Bar (thin rectangle on lifeline)** A narrow rectangle placed on the lifeline showing when an object is actively processing — i.e. when it is doing work in response to a message.
- 4. Messages (horizontal arrows)** The actual communication between objects.
Synchronous (Solid arrow) → Caller waits for a response before continuing
Return (Dashed arrow) → The response sent back to the caller



SEQUENCE Diagram

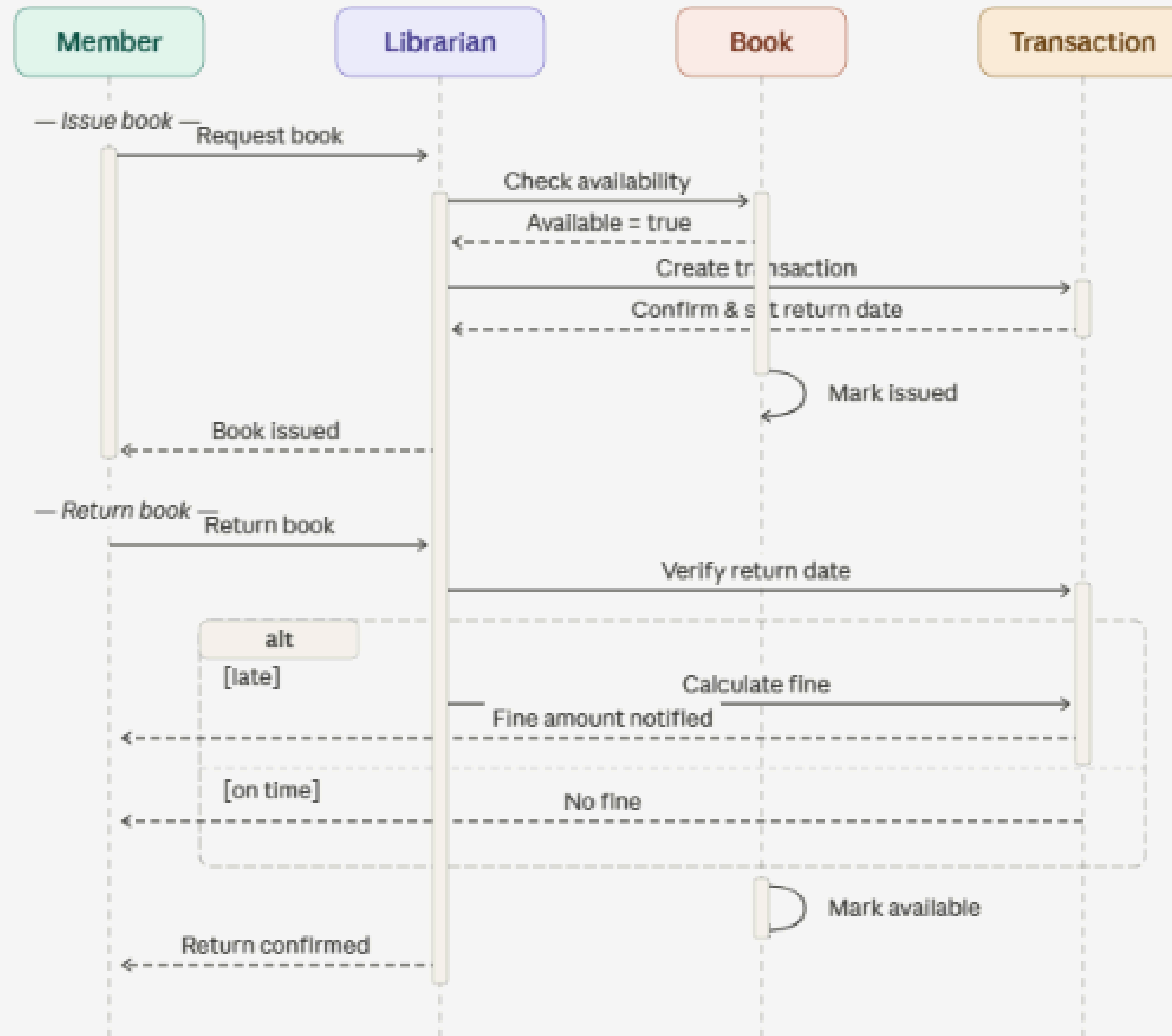
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Sequence diagram - Book issue & return



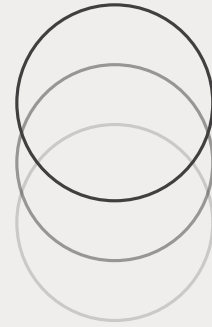


Class Diagram

SUMMARY

A class diagram is a UML structural diagram that shows the blueprint of a system — what classes exist, what data they hold, what they can do, and how they relate to each other. It is drawn before writing code and is the most widely used diagram in object-oriented design.

CLASS Diagram



THE 4 CORE ELEMENTS

1. **Class box** — every class is drawn as a rectangle divided into 3 compartments:

- Top: class name (bold)
- Middle: attributes (data the class stores), prefixed with – (private) or + (public)
- Bottom: methods (operations the class can perform), also prefixed with visibility

2. **Attributes** follow the pattern visibility name: type. Example: – price: Double

3. **Methods** follow the pattern visibility name(params): returnType. Example: + placeOrder(): Order

4. **Visibility markers:**

- + public — accessible from anywhere
- – private — only inside the class itself

5. **Multiplicity labels** on association lines show how many instances relate. Common notations: 1, * (many), 0..1 (zero or one), 1..* (one or more)

6. **Relationships**

Relationship	Between	Type	Meaning
Customer → Order	1 to *	Directed Association	A customer places many orders
Order ◆→ OrderItem	1 to *	Composition	Order items cannot exist without an order
Order ◆→ Payment	1 to 1	Composition	Payment is created by and tied to an order
Restaurant ◇→ MenuItem	1 to *	Aggregation	A restaurant has many menu items; items can exist independently
OrderItem ---→ MenuItem	—	Dependency	OrderItem uses MenuItem to get price and name
Order → DeliveryAgent	1 to 1	Directed Association	An order is assigned to one delivery agent
Restaurant ↔ Order	* to *	Directed Association	A restaurant accepts many orders



CLASS Diagram

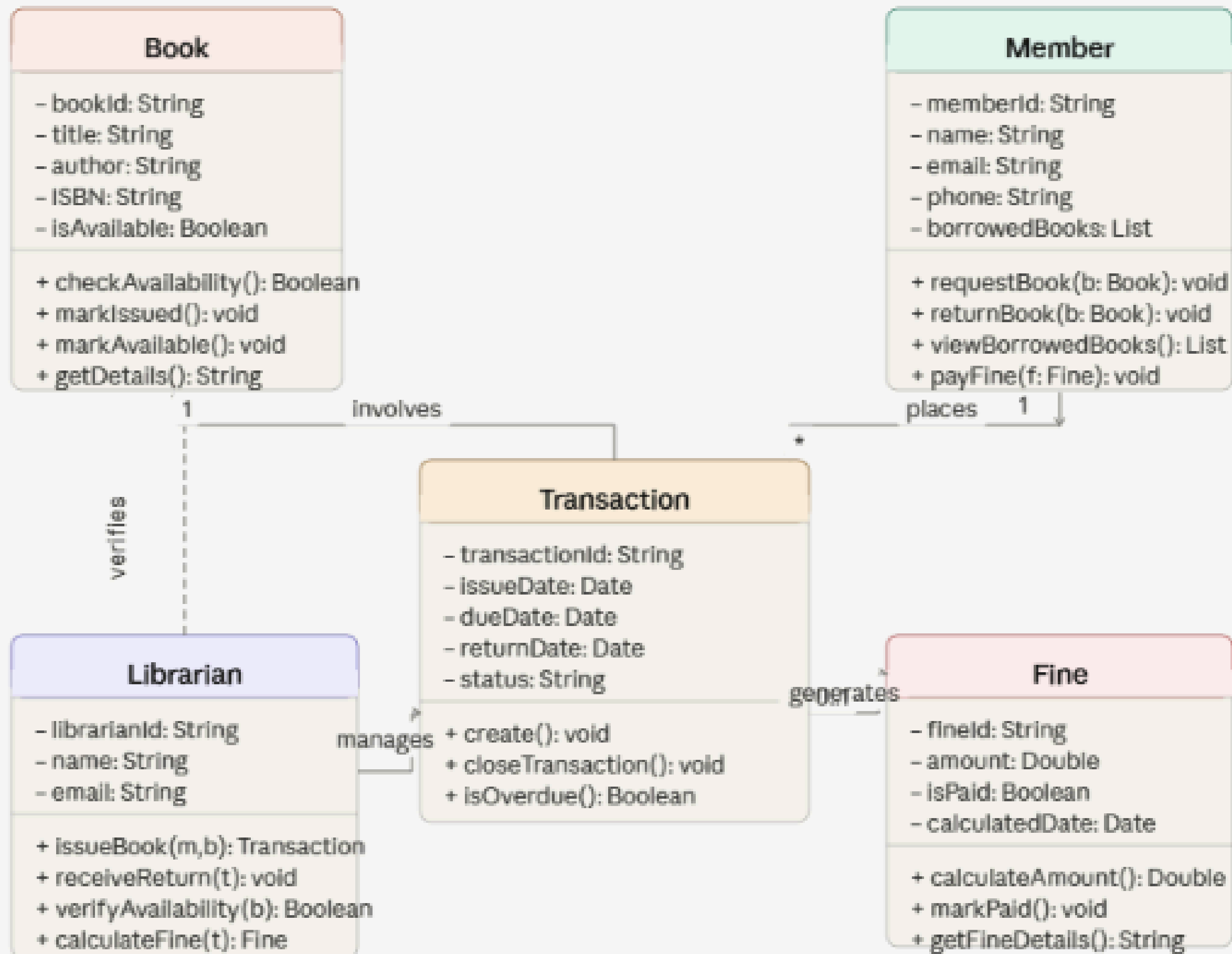
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Class diagram – Library Management System



Identify and explain any 4 types of relationships used in the class diagram

1. Association

A general "uses" or "knows about" relationship. Shown as a plain line (with optional arrow).



A Member places one or more Transactions. Neither owns the other — they simply know about each other.

2. Aggregation

A "has-a" relationship where the part can exist independently. Shown with a hollow diamond at the whole end.



A Librarian manages Transactions. But a Transaction can exist independently (recorded in the system) even if the librarian leaves.

3. Composition

A strong "owns" relationship where the part cannot exist without the whole. Shown with a filled diamond.



A Fine is generated by a Transaction. If the Transaction is deleted, the associated Fine ceases to exist — it has no meaning on its own.

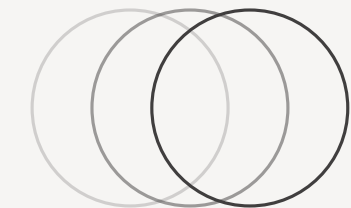
4. Dependency

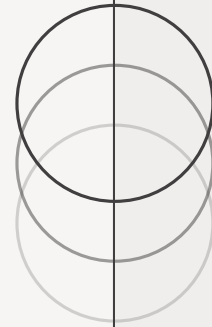
A "uses temporarily" relationship — one class uses another as a parameter or return type. Shown as a dashed arrow.



Librarian uses Book as a parameter in methods like verifyAvailability(b: Book). Librarian does not hold a permanent reference to Book — just uses it when needed.

→ Association ◇→ Aggregation ◆→ Composition -.-.-> Dependency





Homework

IDENTIFY THE OBJECT CLASSES INVOLVED IN **WEATHER STATION** AND DRAW THE CORRESPONDING USE CASE, SEQUENCE & CLASS DIAGRAM.
ALSO IDENTIFY THE RELATIONSHIPS IN THE CLASS DIAGRAM.